MODULE 5A · EPISODE 1

The Earth's Atmosphere: Structures and Layers

From the Weather Below to the Edge of Space

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

The Earth's Atmosphere: Structures and Layers

This episode journeys upward through the thin envelope of gas that makes life on Earth possible. The panel begins with the atmosphere's composition and its four main layers, separated by boundary zones called pauses, where the temperature trend reverses. It then climbs through each layer: the troposphere, where all our weather happens and jet streams blow; the stratosphere, home of the ozone layer; the frigid mesosphere, with its noctilucent clouds, ionosphere, and meteors; and the thermosphere, where the aurorae glow and outer space begins.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist, atmospheric physics

Learning Objectives

After this episode you should be able to:

- Describe the atmosphere's composition and explain why its small amount of carbon dioxide matters.
- Name the four main layers of the atmosphere and the pauses that separate them, and explain what happens to the temperature trend at each pause.
- Explain how the height of the troposphere varies with latitude, and why nearly all of Earth's weather occurs there.
- Describe how temperature and pressure change with height in the troposphere, and give typical values at sea level and at the tropopause.
- Describe jet streams, and explain how aircraft, balloons, birds, and thunderstorm clouds relate to the troposphere and tropopause.
- Explain the stratosphere's temperature inversion, the role of the ozone layer, and the history of the ozone hole.
- Describe the phenomena of the stratosphere and mesosphere, including nacreous clouds, blue jets, noctilucent clouds, airglow, red sprites, and meteors.
- Explain what the ionosphere is and how skywave lets shortwave and ham radio signals travel around the world.
- Explain why the thermosphere is so hot yet would not feel hot, how aurorae form, and where space begins.

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QUICK REFERENCE

Key Terms

Term	Definition
Atmosphere	The gaseous envelope that encapsulates the Earth; a dynamic, life-sustaining system that has evolved over billions of years.
Greenhouse effect	The warming of Earth by gases, such as carbon dioxide, that trap heat; it helps regulate Earth's temperature.
Pause	A relatively thin boundary zone between two layers of the atmosphere, where the temperature trend reverses.
Troposphere	The first and lowest layer of the atmosphere, closest to the surface, where virtually all weather occurs.
Tropopause	The top of the troposphere, where the air stops cooling with height.
One atmosphere	The air pressure at sea level: 101,325 pascals, or 14.7 pounds per square inch.
Jet stream	A high-speed, west-to-east flow of wind found just below the tropopause.
Anviling out	The spreading of a thunderstorm cloud into a flat top when it stops rising at the tropopause.
Cumulonimbus cloud	A towering thunderstorm cloud.
Stratosphere	The second layer of the atmosphere, above the tropopause, where temperature rises (or stays the same) with height.
Stratopause	The top of the stratosphere, at about 50 kilometers at mid-latitudes.
Temperature inversion	A region where temperature rises with height instead of falling.
Ozone layer	A layer in the stratosphere, mostly 15 to 35 kilometers up, where ozone absorbs harmful ultraviolet radiation.
Chlorofluorocarbons (CFCs)	Human-made compounds that caused depletion of the ozone layer and are now banned internationally.
Nacreous clouds	Mother-of-pearl clouds: tiny ice crystals in the polar winter stratosphere that shimmer with iridescent colors.
Blue jets	Lightning-related events that shoot upward from the tops of thunderstorms into the stratosphere.
Mesosphere	The third layer of the atmosphere, above the stratopause, which contains the coldest temperatures on or near Earth.
Mesopause	The top of the mesosphere, at about 85 kilometers at mid-latitudes.
Noctilucent clouds	Clouds of tiny ice crystals in the summer mesosphere that glow with reflected sunlight during deep twilight.
Ionosphere	An irregularly shaped region of electrically charged gas, from about 60 to 600 kilometers up.

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Term	Definition
Plasma	A gas of free electrons and positive ions.
Skywave (skip)	The reflection of radio waves from 3 to 30 million hertz off the ionosphere, back to Earth.
Airglow	A faint, often greenish glow given off by the whole night sky, as atoms and molecules energized by daytime sunlight release that energy.
Red sprites	Lightning-related flashes in the mesosphere, high above thunderstorms, at the moment of powerful cloud-to-ground strikes.
Laser guide star	Sodium atoms near the top of the mesosphere, made to glow by a laser, used to correct a telescope's images.
Meteoroid	A typically small chunk of space rock; it glows as a meteor when it plows through the atmosphere.
Thermosphere	The fourth and final true layer of the atmosphere, from the mesopause to the thermopause.
Thermopause	The top of the thermosphere, at a variable but typically maximum height of 600 kilometers.
Exosphere	The extremely thin outermost region, above the thermopause, that fades gradually into space.
Magnetosphere	Earth's protective magnetic field, which traps charged particles from the solar wind.
Aurorae	The northern and southern lights, produced when charged particles collide with air in the thermosphere.
Kármán line	The altitude of 100 kilometers, widely recognized as the beginning of outer space.
Anacoustic zone	The region, beginning at about 160 kilometers, where the air is too thin to carry ordinary sound.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
78%, 21%, 0.93%, 0.04%	Nitrogen, oxygen, argon, and carbon dioxide in the atmosphere, excluding water vapor
18 km (11 mi)	Height of the troposphere at the equator, about 59,000 feet
11 km (7 mi); 8 km	Height of the troposphere at mid-latitudes (about 36,000 feet); over the poles
~75% / 99%	The troposphere's share of Earth's air by mass / of its water vapor
15°C (59°F)	Average air temperature at sea level worldwide

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Value	What it refers to
−57°C (−70°F)	Average temperature at the tropopause at mid-latitudes
101,325 Pa (14.7 psi)	Air pressure at sea level: one atmosphere
28,000–45,000 ft	Typical cruising altitudes of commercial jets at mid-latitudes
30° N; 50–60° N	Latitudes of the subtropical and polar jet streams
80–140 mph	Typical jet stream speeds (129–225 km/h)
~69,000 ft	Hot air balloon altitude record
37,000 ft	Highest-flying bird on record, Rüppell's griffon vulture
35,000–56,000 ft	Height of thunderstorm anvil tops, mid-latitudes to tropics
50 km (31 mi)	The stratopause at mid-latitudes, about 164,000 feet
99%	Share of Earth's air by mass in the troposphere and stratosphere together
−3°C (27°F)	Temperature at the stratopause
15–35 km; 20–25 km	Extent of the ozone layer; height of peak ozone abundance
~1/1,000	Air pressure at the stratopause, compared with sea level
60,000–70,000 ft	Operating altitudes of the U-2 spy plane
85,000 ft	Highest altitude of the SR-71 Blackbird
~136,000 ft	Alan Eustace's record skydive from a balloon
85 km (53 mi)	The mesopause at mid-latitudes, about 280,000 feet
−90°C (−130°F)	Average temperature at the mesopause
below −130°C (−200°F)	Mesopause temperature over the summer poles
60–600 km	Extent of the ionosphere
3–30 MHz	Radio frequencies that skywave works for (3 to 30 million hertz)
600 km (373 mi)	Typical maximum height of the thermopause
2,000°C (3,632°F)	Maximum thermosphere temperature, high sunspot activity
500°C (932°F)	Maximum thermosphere temperature, low sunspot activity
~300 km	Where the ionosphere's electrons are most densely packed
100–500 km	Altitudes where aurorae glow
100 km	The Kármán line, the beginning of outer space
160 km	Where the anacoustic zone begins
400 km (250 mi)	Orbit of the International Space Station

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Composition and Structure

As far as we know, Earth is the only “rock” in the entire Universe that supports life, and its atmosphere is central to that uniqueness. We can define the atmosphere as the **gaseous envelope that encapsulates the Earth**, but it is much more than a layer of gas: it is a dynamic, life-sustaining system that has evolved over billions of years.

Excluding water vapor, the atmosphere consists primarily of four gases: **78% nitrogen, 21% oxygen, 0.93% argon, and only about 0.04% carbon dioxide**. Small as it is, that carbon dioxide plays a crucial role in regulating Earth's temperature through the **greenhouse effect**.

The atmosphere is divided into **four main layers**, separated by relatively thin boundary zones called **pauses**, where the temperature trend reverses. Each layer has its own characteristics, temperature profile, and phenomena.

Part 2 — The Troposphere

The **troposphere** is the first layer, closest to the surface where we live and breathe. Its height varies with latitude: about **18 kilometers** (11 miles, 59,000 feet) at the equator, about **11 kilometers** (7 miles, 36,000 feet) at mid-latitudes, and only about **8 kilometers** over the poles.

Though it is the thinnest layer, the troposphere is incredibly dense with matter. It holds about **75% of Earth's air by mass** and **99% of the atmosphere's water vapor**, which is why virtually all of Earth's weather occurs there: weather needs water in its various phases.

Temperature **decreases with height** in the troposphere, from an average of **15°C (59°F)** at sea level to about **−57°C (−70°F)** at the top, the **tropopause**, at mid-latitudes. Above the tropopause, the air stops cooling and eventually starts to warm again. Sea-level pressure is **101,325 pascals**, or 14.7 pounds per square inch, which we call **one atmosphere**. Both temperature and pressure fall dramatically with height.

Part 3 — Flying Through the Troposphere

At mid-latitudes, commercial jets cruise at about **28,000 to 45,000 feet**, in the upper troposphere and, on the highest flights, just into the lower stratosphere. The retired **Concorde** cruised at around 60,000 feet, well above the tropopause, avoiding much of the weather and turbulence below.

Just below the tropopause flow the **jet streams**, high-speed winds blowing from west to east. In the Northern Hemisphere, the **subtropical jet stream** lies near 30° N, and the **polar jet stream** wanders at about 50 to 60° N. Their winds typically reach **80 to 140 miles per hour** (129 to 225 km/h), and eastbound airliners ride them as tailwinds to save time and fuel.

Most hot air balloons stay far below the jet streams, though record-setting round-the-world balloon flights have ridden them, and one hot air balloon has reached a record of about **69,000 feet**. The highest-flying bird on record, **Rüppell's griffon vulture**, has been observed at

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37,000 feet over Africa.

Because the temperature trend reverses at the tropopause, towering **cumulonimbus** thunderstorm clouds usually stop rising there and spread into a flat top, from about 35,000 feet at mid-latitudes to about 56,000 feet in the tropics. This is called **“anviling out.”**

Part 4 — The Stratosphere

The **stratosphere** is the second layer, extending at mid-latitudes from the tropopause to about **50 kilometers** (31 miles, 164,000 feet), the **stratopause**. Together, the troposphere and stratosphere hold **99% of Earth's air by mass**.

Counterintuitively, temperature **rises (or stays the same)** with height in the stratosphere. This **temperature inversion** comes from interactions between ozone, oxygen, and the Sun's high-frequency ultraviolet radiation. The stratopause can reach a comparatively balmy **−3°C (27°F)**.

The **ozone layer** lies mostly between **15 and 35 kilometers**, with peak abundance at 20 to 25 kilometers. Along with oxygen, it absorbs the most destructive wavelengths of the Sun's ultraviolet radiation, which would cause serious DNA damage if it reached the surface. Human release of **chlorofluorocarbons (CFCs)** produced an ozone hole over Antarctica; an international ban has let the layer slowly start healing. Near the surface, ozone is a pollutant that can impair lung function.

In the polar winter, water vapor can freeze into tiny ice crystals that form **nacreous**, or mother-of-pearl, **clouds**, which shimmer during civil twilight. **Blue jets** shoot upward from thunderstorm tops into the stratosphere. At the stratopause, the air pressure is only about **one thousandth** of sea-level pressure, far too thin for jet engines. The **U-2** spy plane operated at 60,000 to 70,000 feet and the **SR-71 Blackbird** flew as high as 85,000 feet, while **Alan Eustace** sky-dived from a helium-filled balloon at just under 136,000 feet.

Part 5 — The Mesosphere

The **mesosphere** is the third layer, extending at mid-latitudes from the stratopause to about **85 kilometers** (53 miles, 280,000 feet), the **mesopause**. Temperature **decreases** with height, making this the coldest region on or near Earth: about **−90°C (−130°F)** at the mesopause, and below **−130°C (−200°F)** over the summer poles, colder than anywhere on Earth's surface.

There, water vapor, some of it from the breakdown of methane, freezes into tiny ice crystals that form **noctilucent clouds**. Strangely, they appear in summer, glowing with reflected sunlight during deep twilight, long after sunset.

The **ionosphere**, an irregular region of electrically charged gas, begins in the mesosphere at about 60 kilometers and extends to about 600 kilometers. Solar radiation knocks electrons loose from atoms and molecules, creating a **plasma** of free electrons and positive ions that can reflect radio waves. This reflection, **skywave** or **skip**, works for frequencies from **3 to 30 million hertz** and lets shortwave and ham radio operators communicate around the world.

On moonless nights, the whole sky gives off a faint, often greenish **airglow**, as atoms and molecules energized by daytime sunlight slowly release that energy as light. **Red sprites** flash

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high above thunderstorms at the moment of powerful cloud-to-ground lightning strikes.

Near the top of the mesosphere, at about 90 kilometers, a thin layer of **sodium atoms** left by burned-up meteors can be made to glow with a laser. These **laser guide stars** let telescopes continuously adjust flexible mirrors to remove the blurring caused by atmospheric turbulence. And it is in the mesosphere that **meteoroids** burn up: plowing through the thin air at tremendous speeds, they compress and heat the air so intensely that they glow.

Part 6 — The Thermosphere

The **thermosphere** is the fourth and final true layer. It starts at the mesopause and extends to a variable but typically maximum height of **600 kilometers** (373 miles), the **thermopause**. Above it, the extremely thin **exosphere** fades gradually into space.

As in the stratosphere, temperature **rises** with height in the lower thermosphere, and the maximum depends dramatically on sunspot activity: about **2,000°C (3,632°F)** on the sunlit side during high activity, and still about **500°C (932°F)** during low activity. Yet because the air is so thin, you wouldn't actually feel hot there.

The ionosphere's electrons are most densely packed at about **300 kilometers**. Throughout the thermosphere, the few air molecules absorb the Sun's very short-wave ultraviolet and X-ray radiation, which ionizes them and gives them tremendous kinetic energy, producing those extreme temperatures.

The thermosphere is home to the **aurorae**. The **magnetosphere** traps charged particles from the solar wind and funnels some down along magnetic field lines into oval zones around the north and south magnetic poles, where they collide with air **100 to 500 kilometers** up. Green, the most common color, and the deep red seen highest both come from **oxygen** atoms; the blue, violet, and pinkish-red fringe at the bottom comes from **nitrogen** molecules: a giant neon sign in the sky.

The thermosphere also marks boundaries for space exploration. At **100 kilometers**, the **Kármán line**, the atmosphere is so thin that it is widely recognized, including by the European Space Agency, as the beginning of outer space. At about **160 kilometers**, the **anacoustic zone** begins, where ordinary sound can no longer travel. And the **International Space Station** orbits within the thermosphere at about **400 kilometers**, or 250 miles.

From the weather-generating troposphere, through the ozone-rich stratosphere and the frigid mesosphere, to the thermosphere where space begins, each layer plays its part in making Earth the remarkable oasis of life it is.

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TRY IT YOURSELF

Build a scale model. Earth's radius is about 6,371 kilometers. If Earth were a basketball with a radius of 12 centimeters, each millimeter would stand for about 53 kilometers. On that scale, how thick is the troposphere at mid-latitudes (11 km)? Where would the Kármán line (100 km) and the International Space Station (400 km) be?

(Answer: the troposphere would be about 0.2 mm thick, about two sheets of paper. The Kármán line would be about 1.9 mm up and the Space Station about 7.5 mm, less than a third of an inch. All of our weather happens in a layer you could barely feel with a fingernail.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Composition and Structure

1. What makes Earth unique in the Universe, as far as we know? What role does its atmosphere play?

2. How can we define Earth's current atmosphere? Why is it “so much more than just a layer of gas”?

3. Excluding water vapor, which four gases make up most of Earth's atmosphere, and in what proportions?

4. Carbon dioxide makes up only about 0.04% of the atmosphere. Why is it nevertheless important?

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5. How many main layers is the atmosphere divided into? What are the “pauses,” and what happens at them?

Part 2 — The Troposphere

1. Where is the troposphere? How does its height vary with latitude? Give its height at the equator, at mid-latitudes, and over the poles.

2. The troposphere is the thinnest layer, yet it is described as incredibly dense with matter. What shares of Earth's air and water vapor does it contain?

3. Why does virtually all of Earth's weather occur in the troposphere?

4. How does air temperature change with height in the troposphere? Give the average temperature at sea level and at the tropopause at mid-latitudes.

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5. What is the air pressure at sea level, and what do we call this pressure? What happens to pressure as you climb?

6. What happens to the temperature trend above the tropopause?

Part 3 — Flying Through the Troposphere

1. At what altitudes do commercial jets typically cruise at mid-latitudes? How was the Concorde an exception, and what did it gain?

2. What are jet streams, and where are they found? Name the two jet streams of the Northern Hemisphere and give their latitudes.

3. How fast do jet stream winds move? Why do eastbound airliners like to “ride” them?

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4. How do hot air balloons relate to the jet streams? What is the hot air balloon altitude record?

5. What is the highest-flying bird on record? How high has it been observed, and in which layer?

6. Why do thunderstorm clouds usually stop rising at the tropopause? What is “anviling out,” and at what heights does it happen?

Part 4 — The Stratosphere

1. Where is the stratosphere, and how high does it extend at mid-latitudes? What is its upper boundary called?

2. What share of Earth's air by mass do the troposphere and stratosphere hold together?

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3. How does temperature change with height in the stratosphere, and what causes this? How warm can it get at the stratopause?

4. Where is the ozone layer, and where is ozone most abundant? What does the ozone layer protect life from?

5. What caused the ozone hole over Antarctica, and what has allowed the ozone layer to begin healing? Where is ozone considered a pollutant?

6. What are nacreous clouds? When and where do they form, and when can they be seen?

7. What are blue jets?

8. How low is the air pressure at the stratopause? Give the altitudes reached by the U-2, the SR-71 Blackbird, and Alan Eustace.

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Part 5 — The Mesosphere

1. Where is the mesosphere, and how high does it extend at mid-latitudes? What is its upper boundary called?

2. Why is the mesosphere remarkable for its temperatures? Give the average mesopause temperature and the coldest temperatures over the summer poles.

3. What are noctilucent clouds? What are they made of, when do they appear, and when can they be seen?

4. What is the ionosphere? Where does it begin and end, and how does it form?

5. What is skywave, or skip? Which radio frequencies does it work for, and who uses it?

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6. What is airglow, and what causes it?

7. What are red sprites?

8. What is a laser guide star, and how do astronomers use it?

9. What are meteoroids, and why do they glow as they pass through the mesosphere?

Part 6 — The Thermosphere

1. Where does the thermosphere begin and end? What lies above it?

2. How do temperatures in the thermosphere depend on sunspot activity? Give the maximum temperatures during high and low activity.

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3. If the thermosphere is so hot, why wouldn't you feel hot there?

4. What causes the thermosphere's extreme temperatures? Where are the ionosphere's electrons most densely packed?

5. How are aurorae produced? Where on Earth, and at what altitudes, do they occur?

6. Which gases produce the different auroral colors?

7. What is the Kármán line, and at what altitude is it? What begins at about 160 km?

8. At what altitude does the International Space Station orbit, and in which layer?

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9. For each of the four main layers, state whether temperature rises or falls with height, and name one phenomenon found there.

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YOUR NOTES

Notes

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